Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics

2.4 – Parallel Circuits

Unit 2: Electricity & Energy

The question: *last lesson, every component shared the one path round the loop. What changes if the current gets a choice?* A **junction** is a point where a wire splits into two. Today's circuits all have one: that single fact decides both rules below, and both rules are the **opposite** of series.

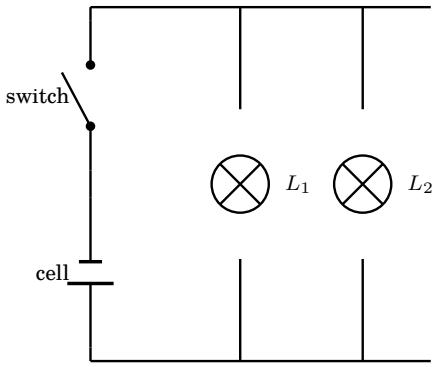
Learning intentions

All students will state and apply the **current rule** and the **voltage rule** for parallel circuits.
Most students will explain why parallel bulbs stay at full brightness however many branches you add.
Some students will calculate a missing current using both rules together.

Part 1 — The current rule

At a junction, the current **splits** between the branches. None of it is lost, so:

The branch currents ADD UP to the total current leaving the cell.



1) **Draw** an ammeter in the branch with L_1, one in the branch with L_2, and one more on the left leg, below the switch, to read the **total** current.

2) **Predict** the current through each bulb, if the total current is **0.6 A** and L_1 and L_2 are identical bulbs.

Build it

Build the circuit and measure the current through L_1, through L_2, and the total current leaving the cell.

$L_1 =$ _____ A $L_2 =$ _____ A total = _____ A

Check: $L_1 + L_2$ should add up close to the total.

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Part 2 — The voltage rule

Both branches connect to the **same** two points — the top bar and the bottom bar. Since voltage is about the difference between two points, and both branches span the *same* two points:

Every branch has the SAME voltage across it — equal to the cell's voltage.

3) **State** the voltage across L_2, if the cell is **9 V** and the voltage across L_1 is **9 V**.

Build it

Measure the voltage across the cell, across L_1, and across L_2.

Cell = _____ V L_1 = _____ V L_2 = _____ V

Check: all three readings should be close to equal.

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Most students — adding a third branch

4) **Explain** what happens to the brightness of L_1 and L_2 if you add a **third** bulb as a new branch, in parallel with the other two. Use the voltage rule in your answer — and say what *does* change, even though the brightness does not.

Some students — using both rules together

5) **Calculate** the current through L_2, if the total current is **0.5 A** and the current through L_1 is **0.35 A**.

Part 3 — On your own

The sockets and lights in a house are wired in **parallel**, not series.

6) **Explain** why this matters in two ways: what happens in the rest of the house if one lamp blows, and what happens to the brightness of the other lamps if you plug in one more appliance. Use the current rule *and* the voltage rule.

HOMEWORK

Make a two-column table, **Series** and **Parallel**. Write one sentence in each column for the current rule, and one sentence in each column for the voltage rule — four sentences in total.